



ME 207: Fluid Dynamics

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Lab Experiment 2: Visualization and Quantification of free and forced Vortices

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Group 14

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Abstract

The aim of this experiment is to visualize and quantify the behavior of these vortices by measuring the free surface for both free and forced vortices.

1 Task 1.1

Q1: What are free and forced vortices? Briefly discuss a few applications/occurrences of such vortices.

A free vortex is a type of motion in which fluid flows in a circular path without any external forces. Also, the torque is zero in free vortex. So, angular momentum ($L = mvr$) is constant.

$$L = mvr = \text{constant}$$

$$v = \frac{k}{r}$$

Where,

- k = proportionality constant,
- r = distance from the axis of rotation.

So In a free cylindrical vortex, the velocity of the fluid decreases when the distance from the rotation axis increases.

In the free vortex, pressure is lowest at the center, and it increases as we go outward. There are no external forces and also there is no external torque is applied to keep the fluid particles in motion. Water drain in the sink, Tornadoes, and hurricanes are examples of free vortex

Forced vortex:

There is a forced vortex when an external force induces a circular motion in a fluid. A rotating impeller, or rotating container can be an example of external force. In the forced vortex, when the water rotates with constant angular velocity, the velocity of fluid particles will also be constant.

$$v = \omega r$$

Where,

- ω = angular velocity
- r = distance from the axis of rotation

In the forced vortex, the entire fluid mass rotates as a solid body which means that every fluid particle will move in a coordinated motion together. Also, pressure increases radially outward. Centrifugal pumps and turbines, and Rotating tanks of water are the example of forced vortex.

Q2: Can you quantify velocity profiles and the associated free surfaces for free and forced vortices?

Free vortex:

There are no external forces in a free vortex. The velocity has an inverse relationship with the distance from the axis of rotation.

$$v = \frac{k}{r}$$

The governing equation for surface profile is derived from the Bernoulli's theorem:

$$\frac{v^2}{2g} + z = C$$

Where,

- $\frac{v^2}{2g}$ = Velocity Head (kinetic energy per unit weight of the fluid),
- z = Potential Head (gravitational potential energy per unit weight of the fluid) or height of the free surface,
- C = Total Energy per Unit Weight.

From these two equations, we get the surface profile of the free vortex which is hyperbolic in nature.

$$C - z = \frac{k^2}{2gr^2}$$

Forced vortex:

Water is forced to rotate in this type of vortex. Assuming the constant angular velocity of the water in our forced vortex. So, the velocity will also be the constant.

$$V = \omega r$$

Where,

- V = velocity of fluid particle,
- ω = angular velocity of fluid particle,
- r = distance from the center.

Using Bernoulli's theorem, we get:

$$\frac{P}{\rho} + \frac{V_{\theta}^2}{2} + z = \text{constant}$$

- P = pressure of the fluid
- ρ = fluid density

- v_θ = tangential velocity
- z = height of the fluid

Let's assume that there is constant atmospheric pressure in this case. Using this two equation, we get the surface profile for the forced vortex. So, the Equation of the forced vortex is:

$$z(r) = C - \frac{\omega^2 r^2}{2}$$

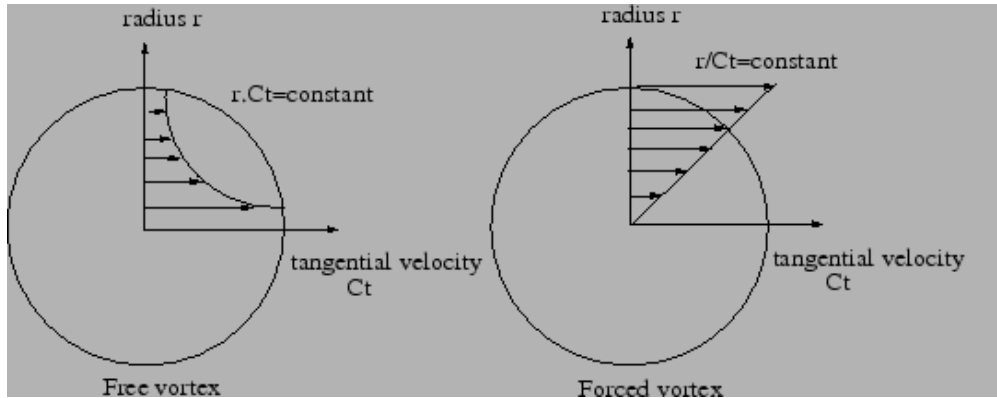


Figure 1: Velocity profile of free vortex (left) and forced vortex (right)

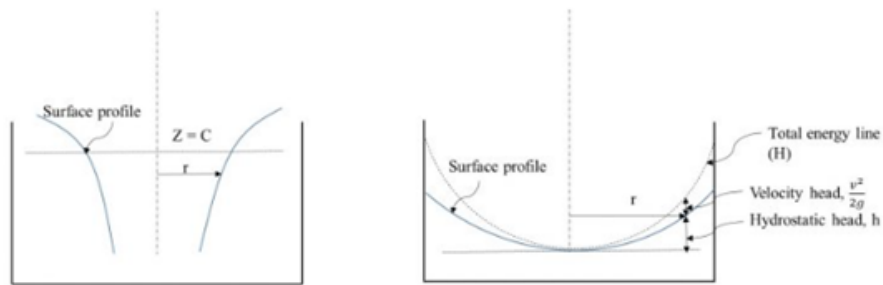


Figure 2: Surface profile of a free vortex (left) and a forced vortex (right)

2 Introduction



Figure 3: Image of the experimental setup.

Vortices can form either naturally or artificially in a controlled environment. They are typically categorized into two types: free vortices and forced vortices. A free vortex occurs, for instance, when water drains from a container through a central opening at the base. No external force is needed to induce rotation in the fluid; the extent of rotation depends on the initial disturbance. Natural occurrences such as whirlpools in rivers and tornadoes are examples of free vortices. In contrast, a forced vortex is generated by external forces acting

on the fluid. This can happen when a container with fluid is mechanically rotated or when the fluid is agitated, such as by paddling. The rotational flow induced by the impellers in a pump, which is observed in turbomachinery, is the common example of forced vortex. Understanding vortex behavior is important for the study natural phenomena and for design-

ing machinery like centrifugal pumps. Both types of vortices are essential in understanding rotational flow dynamics and have applications in fluid machinery, meteorology, and environmental engineering. Engineers have also applied vortex dynamics to create technologies such as hydrodynamic separators for water treatment plants.



Figure 4: Vortex formation.

3 Objectives

- (i) Generate forced and free vortices and visualize the associated free surfaces.
- (ii) Measure the location of the free surface to compare with theoretical estimates.

4 Experimental Procedure

Forced vortex

- Firstly, the cylindrical bowl in which the trajectory was supposed to be measured was filled up to a certain level, and the level of water was noted.
- Then, the bowl was rotated about its central axis, and the frequency of the rotation was measured. decreases in the constant head tank.
- The vortex started appearing, and the bowl was kept rotating for a few minutes, after which the vortex was almost steady.
- Then, the depth of the vortex was measured at different points across the diameter of the bowl, separated by a distance of 3 cm using the rulers attached to the apparatus.
- The entire process was repeated to take three sets of readings.

Free vortex

- We filled the bowl up to a certain level, but here, in free vortex creation, we kept both the inlet and outlet valves open such that the net volume of the water in the bowl approximately remains constant.
- Then, the bowl was rotated again, and the inlet flow was adjusted since the outlet flow reduced as the bowl started rotating.

- From this, we must measure the time until it reaches 5 cm. We note the measured time.
- Once it stabilized, the rotation was stopped, and the depth of the vortex was measured along its diameter for the points located near the center of the vortex separated by a distance of 0.5 cm.
- The entire process was repeated to take seven sets of readings.

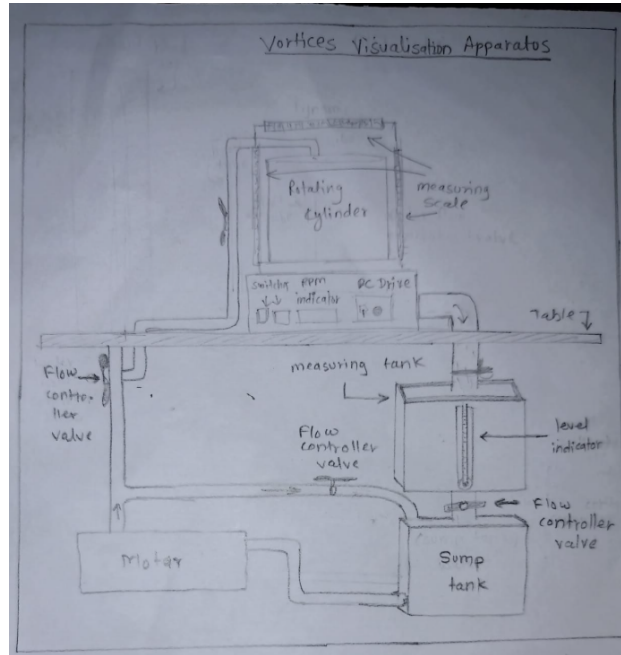


Figure 5: Schematic diagram of the apparatus

5 Results

In the following section we will computing the results obtained for forced vortex and free vortex

The cylinder is rotated at a constant 129-130 rotations per minute.

5.1 Forced Vortex

Firstly, we will be displaying the values obtained experimentally for forced vortex. We took a total of three readings and calculated the shape of curve obtained in case of forced vortex. Please Note that all the three readings were taken by different group members.

In the table given below the x represents the horizontal position of the reading taken, the y represents the vertical position of the reading taken y_{avg} repres represents the average value

of height of three readings taken and is denoted by the formula

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} \implies y_{\text{avg}} = \bar{y} = \frac{y_1 + y_2 + y_3}{3}$$

The standard deviation for the readings has been calculated using the formula

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}} \implies \sigma = \sqrt{\frac{\sum_{i=1}^3 (y_i - \bar{y})^2}{2}}$$

X (cm)	Y1 (cm)	Y2 (cm)	Y3 (cm)	Y Average (cm)	σ (cm)	Final Reading (cm)
-15	23.9	24.1	24.6	24.2	0.36	24.2 ± 0.36
-12	16.3	14.5	15.5	15.43	0.9	15.43 ± 0.9
-9	8.6	6.7	9.8	8.37	1.56	8.37 ± 1.56
-6	4.4	2.6	5.6	4.2	1.51	4.2 ± 1.51
-3	2	1.1	3.3	2.13	1.11	2.13 ± 1.11
0	1.4	1.3	2.4	1.7	0.61	1.7 ± 0.61
3	2	2.3	3.1	2.47	0.57	2.47 ± 0.57
6	4.9	4.8	6	5.23	0.67	5.23 ± 0.67
9	8	10.6	9.7	9.43	1.32	9.43 ± 1.32
12	15.4	18	15	16.13	1.63	16.13 ± 1.63
15	23.9	24.1	24.6	24.2	0.36	24.2 ± 0.36

Table 1: Experimental Data for forced vortex

5.2 Free Vortex

Secondly, we will be displaying the values obtained experimentally for forced vortex. We took multiple readings for free vortex but had to discard several values as the apparatus for the experiment was faulty. Please note that the readings obtained were also taken in the forced vortex apparatus which resulted in very abnormal values, hence the readings cannot be considered correct to plot the final graph accurately.

We have considered three readings to be inserted in the table below The least count for the scale is 0.1cm

X1 (Cm)	Y1(cm)	Final Y reading (cm)
2	5	5 ± 0.1
1.5	5.3	5.3 ± 0.1
1	5	5 ± 0.1
0.5	5.2	5.2 ± 0.1

Table 2: Free vortex reading 1

X1 (Cm)	Y2(cm)	Final Y reading (cm)
2	3.5	3.5 ± 0.1
1.5	2.9	2.9 ± 0.1
1	4.2	4.2 ± 0.1
0.5	4.5	4.5 ± 0.1

Table 3: Free vortex reading 2

X1 (Cm)	Y3(cm)	Final Y reading (cm)
2	4	4 ± 0.1
1.5	3.4	3.4 ± 0.1
1	3.2	3.2 ± 0.1
0.5	2.9	2.9 ± 0.1

Table 4: Free vortex reading 3

Out of all the readings obtained, the one closed to the theoretical values is the reading number - 2, therefore, it is the only value that should be considered for analysis.

6 Error Analysis

In this section, we will compare the experimental results with the theoretical results for the forced vortex and the free vortex

6.1 Forced vortex

Firstly, we will plot the values obtained on the graph; theoretically in a forced vortex the graph will be a parabola, but in real life due to random errors the values obtained can have deviate.

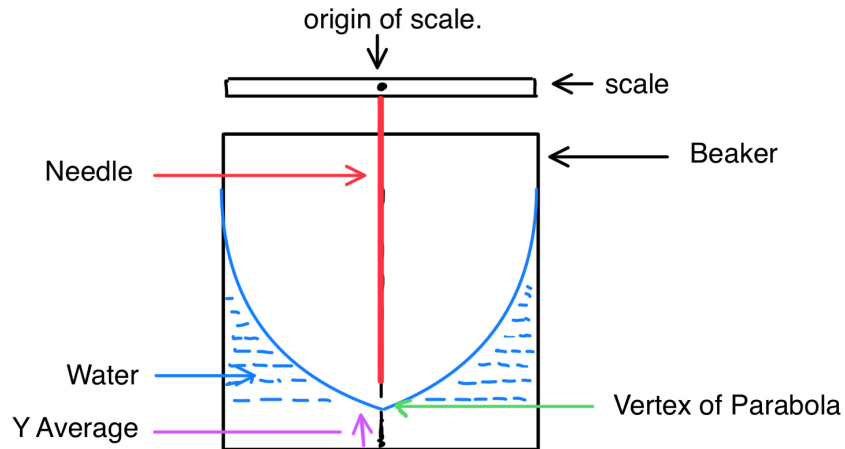


Figure 6: Forced Vortex Figure

The equation of the following parabola is given by:

$$(x - x_o)^2 = 4a(y - y_o)$$

Where:

- x and y are the coordinates of a point on the parabola.
- x_0 is 0 and y_0 is a variable whose value can be calculated.
- a is the distance from the vertex to the focus.
- using two points on the experimental values matching the best fitting parabola we can calculate the equation of parabola.
- The first best fitting point is (-3,2.13).
- The second best fitting point is (12,16.13).

$$\begin{aligned} 12^2 &= 4a(16.13 - y_0) \Rightarrow 144 = 4a(16.13 - y_0) \\ (-3)^2 &= 4a(2.13 - y_0) \Rightarrow 9 = 4a(2.13 - y_0) \end{aligned}$$

Dividing the first equation by the second:

$$\begin{aligned} \frac{144}{9} &= \frac{4a(16.13 - y_0)}{4a(2.13 - y_0)} \\ 16 &= \frac{16.13 - y_0}{2.13 - y_0} \\ 16(2.13 - y_0) &= 16.13 - y_0 \\ 15y_0 &= 17.95 \\ y_0 &\approx 1.1967 \end{aligned}$$

Substituting y_0 back into the second equation:

$$\begin{aligned} 9 &= 4a(2.13 - 1.1967) \\ a &\approx 2.41 \end{aligned}$$

The equation of the parabola is:

$$x^2 = 4(2.41)(y - 1.1967) \Rightarrow x^2 = 9.64(y - 1.1967)$$

The final theoretical equation of the parabola is:

$$\mathbf{x^2 = 9.64(y - 1.2)}$$

The table below shows the comparison of theoretical and experimental values

The average percentage error for the readings taken is **9.1%**

X (cm)	Y avg exp (cm)	Y theoretical (cm)	Percentage error
-15	24.2	24.54	1.39
-12	15.43	16.13	4.34
-9	8.37	9.6	12.81
-6	4.2	4.93	14.81
-3	2.13	2.13	0
0	1.7	1.2	41.67
3	2.47	2.13	15.96
6	5.23	4.93	6.09
9	9.43	9.6	1.77
12	16.13	16.13	0
15	24.2	24.54	1.39

Table 5: Comparison of Experimental and Theoretical Y-values with Percentage Error

The graph below shows the best fitting curve (the parabola) and also contains the graph obtained through experimental readings.

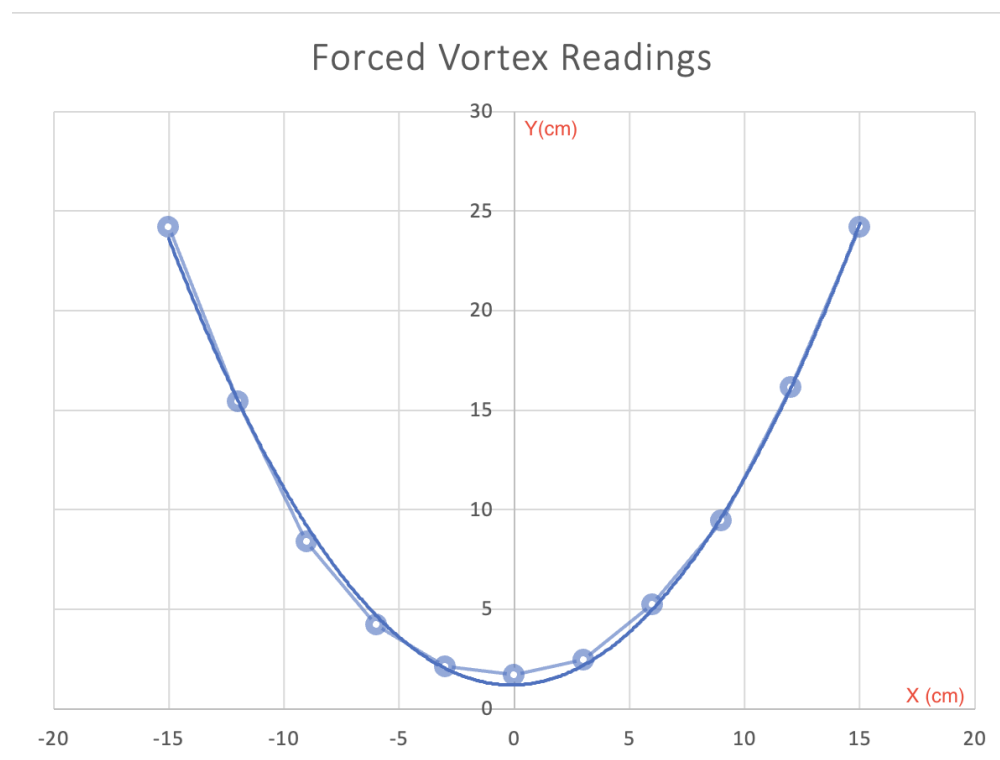


Figure 7: Comparison Graph for Forced Vortex

6.2 Free Vortex

The apparatus used for the free vortex experiment was the same as the one used in forced vortex, which is incorrect but due to technical difficulties we had to perform the experiment on it.

We took multiple readings but only one of them was good enough to be included in the report, hence we have added those values

The values which are considered are tabulated below.

X1 (Cm)	Y1(cm)	Final Y reading (cm)
2	3.5	3.5 ± 0.1
1.5	2.9	2.9 ± 0.1
1	4.2	4.2 ± 0.1
0.5	4.5	4.5 ± 0.1

Table 6: Free vortex reading 3

The expected graph from theoretical concepts should resemble the following image.

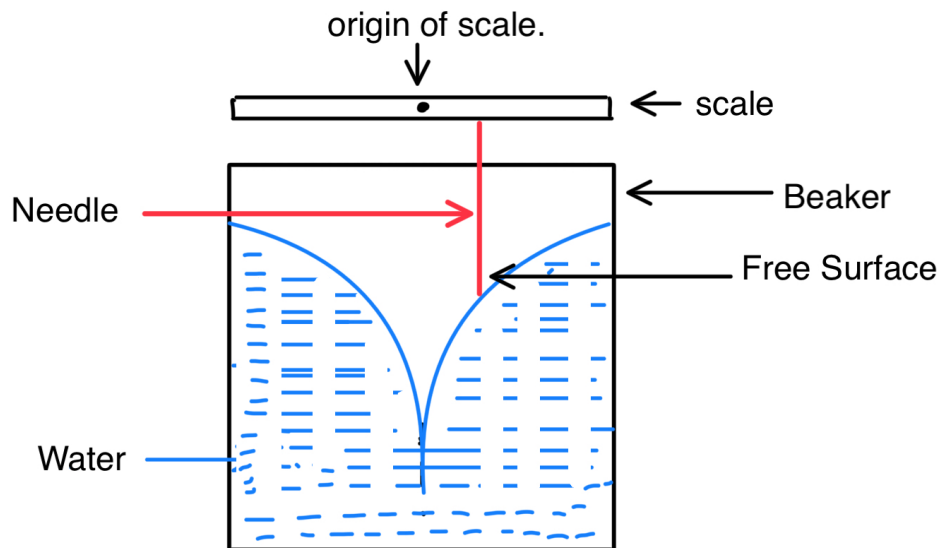


Figure 8: Free Vortex graph (theoretical)

The values that we obtained from the experiment resulted in the following type of graph.

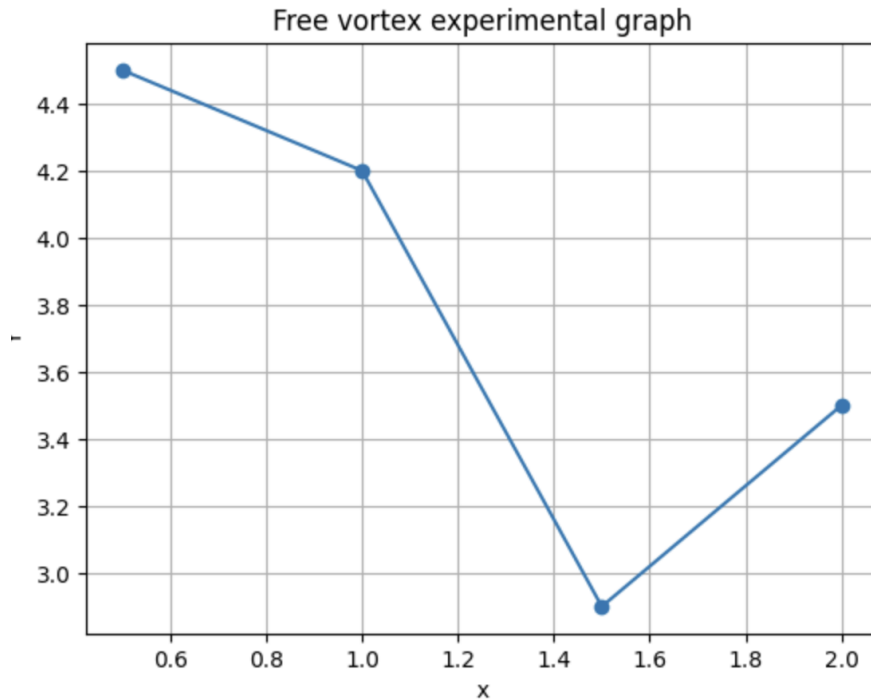


Figure 9: Free Vortex Figure(experimental-2)

This graph depicts the shifting of origin and an exponential nature as the values in x-coordinates decrease.

Due to the large deviations in data observed and lack of available data, we weren't able to do the graph obtained is not correct and due to this we weren't able to do the error analysis for free vortex

Task 2.2:

(a) Measure shape of the free surface for a free vortex at steady state for various orifice sizes (if available).

We have only one specific orifice for this experiment. So, we can find a surface profile for this using the experimental values. It was done in the report in 6.2 on page no. 10-11.

(b) Measure the position of the free surface for the forced vortex for multiple angular velocities of the platform.

We measured the free surface position for the forced vortex for nearly the same angular velocities because the angular velocity in the apparatus fluctuates. The measured values of the free surface position are mentioned in 5.1 on page no. 6-7.

(c) Theoretically compute the free surface shapes for free and forced vortices. Jot down all the assumptions necessary. Now compare your computed free surface shapes to the experimental observed ones. You should quantify experimental uncertainties and errors.

Free vortex:- We can measure the k using the experimental values for the free vortex. If k remains constant, we can measure the free surface shapes for the free vortex. However, the experimental values r and z vary too much. So, we cannot find the free surface shapes for the free vortex. If we do, then it will include lots of errors.

Forced vortex:- We can compute the free surface shapes theoretically using the parabola equation. It was done in the report in 6.1 on page no. 8-9.

Comparison of the theoretical and experimental results:-

Free vortex:- It was done in the report in 6.2 on page no. 10-11.

Forced vortex:- It was done in the report in 6.1 on page no. 8 to 10.

Error Analysis:- It was done in the report in topic 6 on page no. 8-11.

(d) In case there are mismatches between observed and the calculated forces, can you specify some reasons which may cause these mismatches?

Common errors:- The tank is not stable during the rotation, causing an error in the measurement of the vortex depth. The fluid's angular velocity fluctuates, varying the vortices and causing an error in the measuring part. The bottom of the vortex cannot be taken or measured accurately because there is no scale near the bottom. So, we take the approximate values that cause an error.

In the free vortex:- The apparatus of the free vortex experiment has some technical issues or faults. So, we performed the experiment using the apparatus used in the forced vortex experiment. Also, the vortex only appears for 4-5 minutes, and we must take data in this time range. So, it causes errors.

7 Observation

Comparing the values of surface profile using theoretical and experimental methods for the forced vortex. The surface profile calculated using the theoretical method is the same as that found experimentally. In the free vortex experiment, the surface profile calculated using the theoretical method is not the same as that found experimentally.

8 Conclusion

Comparing the surface profile values using theoretical and experimental methods for the forced vortex is nearly the same because it has no significant error and is performed on the correct apparatus. Meanwhile, in the free vortex, the theoretical and experimental values of the surface profile are not the same. There is a large deviation in the data observed because the experiment was performed on an apparatus that was not correct for the experiment and was used for a forced vortex experiment. Also, the free vortex only appears for a short time period of 4-5 minutes.

9 Acknowledgements

We would like to thank our Teaching Assistant Mr. Auromir Saha for explaining the concepts of Free and Forced vortex and also would like to thank our fellow batch mates in helping us understand the process of error analysis

References

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